Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $3x$ - $1, 3, x$ - 3 are the first three terms of an arithmetic progression, then the sum of the first five terms can be

(A) 5

(B) 10

(C) 20

(D) 30

Q.2

Consider the sequence $8\ \mathrm{A}+2\ \mathrm{B}, 6\ \mathrm{A}+\mathrm{B}, 4\ \mathrm{A}$
$2\ \mathrm{A}-\mathrm{B}, \ldots \ldots \ldots$, Which term of this sequence will have a coefficient of A which is twice the coefficient of B?

(A) 10th

(B) 14th

(C) 16th

(D) None of these

Q.3

If $\mathrm{a}=1+(\sqrt{3}-1)+(\sqrt{3}-1)^2+(\sqrt{3}-1)^3+$
.... and $ab = 1$, then a and b are the roots of the equation

(A) $x^2+4x-1=0$

(B) $x^2-4x-1=0$

(C) $x^2+4x+1=0$

(D) $x^2-4x+1=0$

Q.4

Let a_n be the nth term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$
and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is

(A) $\alpha-\beta$

(B) $\beta-\alpha$

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(C) $\frac{\alpha-\beta}{2}$

(D) none

Q.5

If the pth term of an A.P. be $\frac{1}{q}$ and q^{th} term be $\frac{1}{p}$, then the sum of its pq^{th} terms will be

(A) $\frac{pq-1}{2}$

(B) $\frac{1-pq}{2}$

(C) $\frac{pq+1}{2}$

(D) $-\frac{pq+1}{2}$

Q.6

If a, b, c are in A. P., then $(a+2b-c)(2b+c-a)(c+a-b)$ equals

(A) $\frac{1}{2}\mathrm{abc}$

(B) abc

(C) $2\mathrm{abc}$

(D) $4\mathrm{abc}$

Q.7

If, $8, -4$ and 13 be three (not necessarily consecutive term) of an A.P., how many such A.P. s are possible?

(A) 1

(B) 2

(C) infinitely many

(D) no such A.P. is possible

Q.8

If $\log_{10} 2, \log_{10}\left(2^{x}-1\right)$ and $\log_{10}\left(2^{x}+3\right)$ are three consecutive terms of an A.P, then the value of x is

(A) 1

(B) $\log_5 2$

(C) $\log_2 5$

(D) $\log_{10} 5$

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Q.9

What is the 15th term of the series $3, 7, 13, 21, 31, 43, \ldots$?

(A) 205

(B) 225

(C) 238

(D) 241

Q.10

If $a_1, a_2, a_3, \ldots\ldots$ are in A.P. and $a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots\ldots\ldots + a_{2k-1}^2 - a_{2k}^2$ $= \mathrm{M}\left(\mathrm{a}_1^2 - \mathrm{a}_{2\mathrm{k}}^2\right)$. Then $\mathrm{M} =$

(A) $\frac{\mathrm{k}-1}{\mathrm{k}+1}$

(B) $\frac{\mathrm{k}}{2\mathrm{k}-1}$

(C) $\frac{\mathrm{k}+1}{2\mathrm{k}+1}$

(D) none

Q.11

For $-\pi < x < \pi$, the values of x which satisfy the relation $11^{1+|\cos x|+\cos^2 x+\cos^3 x|+\cdots}$ upto $\infty = 121$ are given by

(A) $\pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

(B) $\frac{\pi}{3}, \frac{2\pi}{4}$

(C) $\frac{\pi}{4}, \frac{3\pi}{4}$

(D) None of these

Q.12

If the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P., then their common difference will be :

(A) ± 1

(B) ± 2

(C) ± 3

(D) ± 4

Q.13

Let a, b, c be in AP. Consider the following statements:
1. $\frac{1}{ab}, \frac{1}{ca}$ and $\frac{1}{bc}$ are in AP.

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2. $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}$ and $\frac{1}{\sqrt{a}+\sqrt{b}}$ arein AP.
Which of the statements given above is/are correct?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.14

If the sum and product of the first three terms in an $A.P.$ are 33 and 1155, respectively, then a value of its 11th term is:

(A) -35

(B) 25

(C) -25

(D) -36

Q.15

If $a,\ \ b,\ \ c$ are in A.P., b, c, d are in G.P., and $c,\ \ d,\ \ e$ are in H.P. then $a,\ \ c,\ \ e$ are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) None of these

Q.16

If the angles $\mathrm{A} < \mathrm{B} < \mathrm{C}$ of a triangle are in $\mathrm{A.P.}$ then

(A) $\mathrm{c}^2 = \mathrm{a}^2 + \mathrm{b}^2 - \mathrm{ab}$

(B) $b^2 = a^2 + c^2 - ac$

(C) $c^2 = a^2 + b^2$

(D) None of these

Q.17

Let a, b, c be in A.P. and $|a| < 1, |b| < 1, |c| < 1$. If
$x = 1 + a + a^2 + \ldots\ldots$ to ∞
$y = 1 + b + b^2 + \ldots\ldots$ to ∞
$z = 1 + c + c^2 + \ldots\ldots$ to ∞, then x, y, z are in

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(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Q.18

The angles of a triangle are in A.P. and the radian measure of the smallest to the degree measure of the mean is $\pi : 200$. The greatest angle in radians is

(A) $\frac{11\pi}{30}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) none of these

Q.19

If a, b, c, d are positive real numbers such that $a+b+c+d=2$, then $M=(a+b)(c+d)$ satisfies the relation

(A) $0 < M \leq 1$

(B) $1 \leq M \leq 2$

(C) $2 \leq M \leq 3$

(D) $3 \leq M \leq 4$

Q.20

The sum of each of two sets of three terms in A.P. is 15 . The common difference of the first set is greater than that of the second by 1 and the ratio of the products of the terms in the first set and that of the second set is $7 : 8$. The ratio of the smallest terms in two sets of terms is

(A) $\frac{3}{4}$ or $\frac{11}{12}$

(B) $\frac{2}{3}$ or $\frac{1}{2}$

(C) $\frac{2}{3}$ or $\frac{3}{4}$

(D) None of these

Answers & Solutions

Q.1 Answer:

5

Solution:

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Case-I : $x < \frac{1}{3} \Rightarrow -3x+1, 3, -x+3$ **are in** $A.P.$

$\Rightarrow 6 = -3x+1-x+3$

$\Rightarrow 4x = -2 \Rightarrow x = -\frac{1}{2}$

$\Rightarrow$ **terms are** $\frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}$

$\Rightarrow$ **sum** $= \frac{35}{2}$

Case-II: $\frac{1}{3} \leq x < 3$

$3x-1, 3, -x+3$ **are in A.P.**

$\Rightarrow 6 = 3x-1-x+3 = 2x+2 \Rightarrow x = 2$

$\Rightarrow$ **terms are** $5, 3, 1, -1, -3$

$\Rightarrow$ **sum** $= 5$

Case-III: $x \geq 3$

$\Rightarrow$ **terms are** $3x-1, 3, x-3$

$\Rightarrow 6 = 3x-1+x-3 = 4x-4$

$\Rightarrow 4x = 10 \Rightarrow x = \frac{5}{2}$ **not possible**

Q.2 Answer:

None of these

Solution:

Given sequence $8A+2B, 6A+B, 4A, 2A-B, \ldots\ldots$

This is an AP with common difference $= -2\,A - B$

First term, $a = 8\,A + 2\,B$

n^{th} term of $AP, t_n = a + (n-1)d$

$= 8A + 2B + (n-1)(-2A-B)$

$= 8A + 2B - 2An - Bn + 2A + B$

$= 10A + 3B - 2An - Bn$

$= (10-2n)A + B(3-n)$

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Coefficient of A = twice the coefficient of B

$$=> 10 - 2n = 2(3 - n)$$

$$=> 10 - 2n = 6 - 2n$$

$$=> 10 = 6$$

This is not possible.

$\Rightarrow$ no value of n exists for the given condition.

There is no value of n for which coefficient of A is twice the coefficient of B.

Q.3 Answer:

$x^2 - 4x + 1 = 0$

Solution:

$a = \frac{1}{1-(\sqrt{3}-1)} = \frac{1}{2-\sqrt{3}} = 2 + \sqrt{3}$
and $ab = 1 \Rightarrow b = 2 - \sqrt{3}$
so, a and b **are roots of** $x^2 - 4x + 1 = 0$

Q.4 Answer:

none

Solution:

Let d **be the common difference of the A.P.**
Then $\mathrm{a}_{2\mathrm{r}} = \mathrm{a}_{2\mathrm{r}-1} + \mathrm{d}$
$\therefore \sum_{\mathrm{r}=1}^{100} \mathrm{a}_{2\mathrm{r}} = \sum_{\mathrm{r}=1}^{100} (\mathrm{a}_{2\mathrm{r}-1} + \mathrm{d}) = \sum_{\mathrm{r}=1}^{100} \mathrm{a}_{2\mathrm{r}-1} + 100\ \mathrm{d}$
$\Rightarrow \alpha = \beta + 100\ \mathrm{d} \Rightarrow \mathrm{d} = \frac{\alpha - \beta}{100}$

Q.5 Answer:

$\frac{pq+1}{2}$

Solution:

Since $T_p = a + (p-1)d = \frac{1}{q}$
...
and $T_q = a + (q-1)d = \frac{1}{p}$ ..
From (i) and (ii), we get $a = \frac{1}{pq}$ **and** $d = \frac{1}{pq}$ **Now sum of pq terms** $= \frac{pq}{2}\left[\frac{2}{pq} + (pq-1)\frac{1}{pq}\right]$
$= \frac{pq}{2} \cdot \frac{2}{pq}\left[1 + \frac{1}{2}(pq-1)\right] = \left[\frac{2+pq-1}{2}\right] = \frac{pq+1}{2}$

Q.6 Answer:

4abc

Solution:

$2\ \mathrm{b} = \mathrm{a} + \mathrm{c}$, **so the given expression is** $(a + a + c - c)(a + c + c - a)(2b - b) = 4abc$

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Q.7 Answer:

infinitely many

Solution:

Since an A.P. is either increasing or decreasing, if possible let -4 be the first term of an A.P., whose m^{th} and n^{th} terms are respectively 8 and 13 . Then $8=-4+(\mathrm{m}-1) \mathrm{d}$ and $13=-4+(\mathrm{n}-1) \mathrm{d}$

$\Rightarrow \frac{12}{m-1}=\frac{17}{n-1}=d$

Let $\frac{\mathrm{m}-1}{12}=\frac{\mathrm{n}-1}{17}=\mathrm{k}$, then $\mathrm{m}=12 \mathrm{k}+1$, and n

$=17 \mathrm{k}+1$

$\therefore$ for $\mathrm{k}=1,2,3, \ldots \ldots \ldots$ we get different pairs of values of m and n, which shows that infinite number of A.P's can be obtained

Q.8 Answer:

$\log_2 5$

Solution:

$\log_{10} 2, \log_{10}\left(2^x-1\right)$ **and** $\log_{10}\left(2^x+3\right)$ **are**

in A.P. Hence, common difference will be same.

$\therefore \quad \log_{10}\left(2^x-1\right)-\log_{10} 2$

$=\log \left(2^x+3\right)-\log_{10}\left(2^x-1\right)$

$\therefore \quad \log_{10}\left(\frac{2^{\mathrm{x}}-1}{2}\right)=\log_{10}\left(\frac{2^{\mathrm{x}}+3}{2^{\mathrm{x}}-1}\right)$

$\Rightarrow \quad \frac{2^x-1}{2}=\frac{2^x+3}{2^x-1}$

$\left(2^x-1\right)^2=2\left(2^x+3\right)$

$2^{2 x}-2^{x+1}+1=2^{x+1}+6$

$2^{2 x}-2^{x+2}=5$

Let $2^{\mathrm{x}}=\mathrm{y}$, then $y^2-4 y-5=0$

$y^2-5 y+y-5=0$

$y(y-5)+1(y-5)=0$

$y=-1, y=5$

Therefore, $2^{\mathrm{x}}=5 \mathrm{x}=\log_2 5$

Q.9 Answer:

241

Solution:

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$S = 3 + 7 + 13 + 21 + 31 + \ldots\ldots\ldots + a_n (i)$

$-S = -3 - 7 - 13 - 21 - \ldots\ldots\ldots - a_{n-1} - a_n (ii)$

(i) + (ii)

$\Rightarrow 0 = 3 + 4 + 6 + 8 + 10 + 12 \ldots - a_n$

$\Rightarrow a_n = 3 + 4 + 6 + 8 + 10 + 12 \ldots (n-1)$ terms

Here $4 + 6 + 8 + 10 + 12$... is an AP with $a = 4$ and $d = 2$

Sum of n terms of AP $= (n/2)(2a + (n-1)d)$

So $a_n = 3 + [(n-1)/2](8 + (n-2)2)$

$= 3 + [(n-1)/2](8 + 2n - 4)$

$= 3 + [(n-1)/2](4 + 2n)$

$= 3 + (n-1)(2 + n)$

So 15 th term $= a_{15} = 3 + (15-1)(2 + 15)$

$= 3 + 14 \times 17$

$= 3 + 238$

$= 241$

So the 15 th term of the series $3, 7, 13, 21, 31, 43, \ldots$ is 241.

Q.10 Answer:

$\frac{k}{2k-1}$

Solution:

We have, $a_2 - a_1 = a_3 - a_2 = \ldots\ldots\ldots\ldots\ldots a_{2k} - a_{2k-1} = d$

Hence,

$a_1^2 - a_2^2 = (a_1 - a_2)(a_1 + a_2) = -d(a_1 + a_2)$

$a_3^2 - a_4^2 = (a_3 - a_4)(a_3 + a_4) = -d(a_3 + a_4)$

$a_{2k-1}^2 - a_{2k}^2 = (a_{2k-1} - a_{2k})(a_{2k-1} + a_{2k})$

$= -d(a_{2k-1} + a_{2k})$

Adding, we get $a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots\ldots\ldots + a_{2k-1}^2 - a_{2k}^2$

$= -d(a_1 + a_2 + a_3 + a_4 + \ldots\ldots a_{2k-1} + a_{2k})$

$= -d \cdot \frac{2k}{2}(a_1 + a_{2k}) = -dk(a_1 + a_{2k})$

But $a_{2k} = a_1 + (2k-1)d \Rightarrow -d = \frac{a_1 - a_{2k}}{2k-1}$

$\therefore$ The required sum $= \frac{k}{2k-1}(a_1^2 - a_{2k}^2)$

$\Rightarrow M = \frac{k}{2k-1}$

Q.11 Answer:

$\pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

Solution:

Since, $0 < x < \pi, -1 < \cos x < 1 \Rightarrow 0 \leq |\cos x| < 1.$

We can write the given expression as $11^{1/(1-|\cos x|)} = 121$

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$\Rightarrow \quad \frac{1}{1-|\cos x|} = 2$

$\Rightarrow \quad 1 - |\cos x| = \frac{1}{2}$

$\Rightarrow \quad |\cos x| = \frac{1}{2}$

$\Rightarrow \quad \cos x = \pm\frac{1}{2}$

$\Rightarrow \quad x = \pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

Q.12 Answer:

± 3

Solution:

Roots of Given equation $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P. Let $\alpha - \beta, \alpha, \alpha + \beta$ be the roots of the equation. Sum of the roots $= \alpha - \beta + \alpha + \alpha + \beta =$
$\frac{-(-12)}{1} = 12$

$3\alpha = 12 \Rightarrow \alpha = 4$ **and** $(\alpha - \beta)\alpha + \alpha(\alpha + \beta) + (\alpha + \beta)$

$(\alpha - \beta) = 39$

$\Rightarrow \alpha^2 - \alpha\beta + \alpha^2 + \alpha\beta + \alpha^2 - \beta^2 = 39$

$\Rightarrow 3\alpha^2 - \beta^2 = 39 \Rightarrow 3(4)^2 - \beta^2 = 39$

$\Rightarrow 48 - \beta^2 = 39 \Rightarrow -\beta^2 = 39 - 48 \Rightarrow -\beta^2 = -9$

$\Rightarrow \beta = \pm 3$

Q.13 Answer:

Both 1 and 2

Solution:

Let $\frac{1}{ab}, \frac{1}{ca}, \frac{1}{bc}$ **are in AP.**

$\Rightarrow \frac{1}{ca} - \frac{1}{ab} = \frac{1}{bc} - \frac{1}{ca}$

$\Rightarrow \frac{1}{a}\left(\frac{1}{c} - \frac{1}{b}\right) = \frac{1}{c}\left(\frac{1}{b} - \frac{1}{a}\right)$

$\Rightarrow \frac{b-c}{abc} = \frac{a-b}{abc}$

$\Rightarrow b - c = a - b \Rightarrow 2b = a + c$

$\Rightarrow$ a, b, c **are in** AP. **Which is true**

Now, $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ **are in A.P.**

$\frac{2}{\sqrt{c}+\sqrt{a}} = \frac{1}{\sqrt{b}+\sqrt{c}} + \frac{1}{\sqrt{a}+\sqrt{b}}$

$\Rightarrow 2(\sqrt{b} + \sqrt{c})(\sqrt{a} + \sqrt{b})$

$= (\sqrt{c} + \sqrt{a})(\sqrt{a} + 2\sqrt{b} + \sqrt{c})$

$\Rightarrow 2(\sqrt{ab} + b + \sqrt{ac} + \sqrt{bc})$

$= \sqrt{ac} + 2\sqrt{bc} + c + a + 2\sqrt{ab} + \sqrt{ac}$

$\Rightarrow 2\sqrt{ab} + 2b + 2\sqrt{ac} + 2\sqrt{bc}$

$= 2\sqrt{ac} + 2\sqrt{bc} + 2\sqrt{ab} + c + a$

$\Rightarrow \quad 2b = a + c$

$\Rightarrow$ a, b, c **are in A.P. Which is true.**

Hence, both the statements are correct.

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Q.14 Answer:

-25

Solution:

Let, the first 3 **terms of the** $A.P.$ **are** $a - d,\ a,\ a + d$

$\Rightarrow \text{sum} = 3a = 33 \Rightarrow a = 11$

And product $a - d\ a\ a + d = aa^2 - d^2 = 1155$

$\Rightarrow 1111^2 - d^2 = 1155$

$\Rightarrow 121 - d^2 = \frac{1155}{11} = 105$

$\Rightarrow d^2 = 121 - 105 = 16$

$\Rightarrow d = \pm 4$

For $a = 11,\quad d = 4,$ **the** $A.P.$ **is** $7,\ 11,\ 15,\ \ldots$

The nth **term of an** $A.P.,$ **with first term** A **and common difference** d **is** $T_n = A + n - 1d$

So, $T_{11} = 7 + 10 \times 4 = 47$

For $a = 11,\quad d = -4,$ **the** $A.P.$ **is** $15,\ 11,\ 7,\ \ldots$

So, $T_{11} = 15 + 10\text{-}4 = -25.$

Q.15 Answer:

G.P.

Solution:

$a, b, c \text{in} A.P. \Rightarrow a + c = 2\quad b;\quad b, c, d \text{in} GP$

$\Rightarrow bd = c^2;\quad c, d, c$ **in** $H.P. \Rightarrow d = \frac{2ce}{c + c}$

$\therefore \frac{a + c}{2} \times \frac{2ce}{c + c} = c^2 \Rightarrow \left(a + c\right)e = \left(c + e\right)c$

$\Rightarrow c^2 = ae$

Therefore, in G.P.

Q.16 Answer:

$b^2 = a^2 + c^2 - ac$

Solution:

$A + C = 2\ B$ **and** $A + B + C = 180°\text{so}, B = 60°$

$\therefore \cos 60° = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow b^2 = a^2 + c^2 - ac$

Q.17 Answer:

H.P.

Solution:

We have, $x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c}$

Now a, b, c **are in A.P.**

$\Rightarrow \quad 1 - a, 1 - b, 1 - c$ **are in A.P.**

$\Rightarrow \quad \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ **are in H.P.** $\Rightarrow x, y, z$ **are in H.P**

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Q.18 Answer:

$\frac{11\pi}{30}$

Solution:

Let the angles in degrees in A.P. be taken as $\alpha-\beta, \alpha, \alpha+\beta$

$\therefore \quad A=\alpha-\beta, B=\alpha, C=\alpha+\beta$

$\therefore \quad A+B+C=180^{\circ} \Rightarrow \alpha-\beta+\alpha+\alpha+\beta=180^{\circ} \Rightarrow 3\alpha=180^{\circ} \quad \Rightarrow \alpha=60^{\circ}$

Now, $\frac{\pi}{200}=\frac{A \text{ in radians}}{B \text{ in degrees}}=\frac{(60-\beta)\frac{\pi}{180}}{60} \Rightarrow \beta=6^{\circ}$

$\therefore$ **Greatest angle** $C=\alpha+\beta=60^{\circ}+6^{\circ}=66^{\circ}$

$=66 \times \frac{\pi}{180}$ radian $=\frac{11}{30}\pi$ **radian**

Q.19 Answer:

$0<M \leq 1$

Solution:

As A.M. $\geq$ **G.M. for positive real numbers, we get**

$\frac{(a+b)+(c+d)}{2} \geq \sqrt{(a+b)(c+d)}$

$\Rightarrow \quad M \leq 1$ **(putting values)**

Also, $(a+b)(c+d)>0 \quad [\square a, b, c, d>0]$

$n<M<1$

Q.20 Answer:

$\frac{3}{4}$ or $\frac{11}{12}$

Solution:

Let the first set of number be $\mathrm{a}-\mathrm{d}, \mathrm{a}, \mathrm{a}+\mathrm{d}$. **Then** $\mathrm{a}-\mathrm{d}+\mathrm{a}+\mathrm{a}+\mathrm{d}=15 \Rightarrow \mathrm{a}=5$

The second set of numbers will be $\mathrm{b}-(\mathrm{d}-1), \mathrm{b}, \mathrm{b}+(\mathrm{d}-1)$. **Again,**

$\mathrm{b}-(\mathrm{d}-1)+\mathrm{b}+\mathrm{b}+(\mathrm{d}-1)=15 \Rightarrow \mathrm{b}=5$

Hence, the sets of numbers are $5-\mathrm{d}, 5, 5+\mathrm{d}$ **and** $6-\mathrm{d}, 5, 4+\mathrm{d}$

Further, from the given condition $\frac{(5-d)5(5+d)}{(6-d)5(4+d)}=\frac{7}{8} \Rightarrow \frac{25-d^2}{24+2d-d^2}=\frac{1}{8}$

$\Rightarrow \mathrm{d}^2+14\,\mathrm{d}-32=0 \Rightarrow \mathrm{d}=2,-16$

$\therefore$ **The two sets are** $3, 5, 7$ **and** $4, 5, 6$ **or** $21, 5, -11$ **and** $22, 5, -12$.

$\therefore$ **Ratio of their smallest term is** $\frac{3}{4}$ **or** $\frac{-11}{-12}$ **i.e.** $\frac{3}{4}$ **or** $\frac{11}{12}$

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